

DSM TUT QUIZ-2

1. Use the laws and postulates of Boolean algebra to prove the following:

(a) $A + BC = (A + B)(A + C)$

(b) $(A + B + C)' = A'B'C'$

2. Simplify the following Boolean expression to the minimum number of literals using Boolean algebraic laws.

$$F = A(A + B') + (A' + C)(A + C) + BC'$$

3. Given the Boolean function

$$F = A'B + AC'$$

prove that

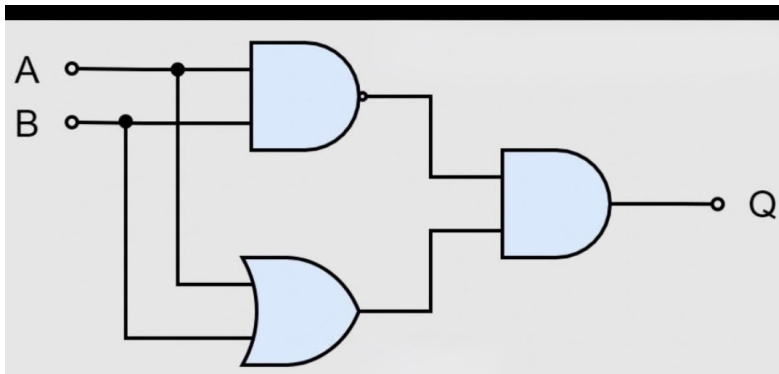
$$F + F' = 1.$$

4. Given the Boolean function

$$F = A(B + C') + B'(C + A)$$

implement the function using only AND, OR, and NOT gates.

5. Consider the following logic-gate representation:



(a) Construct the truth table for the output F .

(b) Express F as a sum of minterms and also represent it using Σm notation.

6. Convert the following Boolean expression from standard form to canonical form:

$$F = (A + B)(A' + C + D')(B + C')$$

7. Given the Boolean function

$$F = (A + B)(A' + C),$$

- (a) Convert F from standard POS form to canonical POS form.
- (b) Convert the canonical POS form obtained in part (a) to canonical SOP form and express the result using Σm notation.

8. Given the Boolean functions

$$F_1(A, B, C) = \Sigma m(1, 3, 5, 7), \quad F_2(A, B, C) = \Sigma m(2, 3, 5, 6),$$

determine:

- (a) $F_1 + F_2$
- (b) $F_1 F_2$

Express your answers using Σm notation. Hence, state the relationship between Boolean OR/AND operations and the union/intersection of minterms.

9. Consider the following truth table for the Boolean function $F(A, B, C)$:

A	B	C	F
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

Find F' in canonical SOP form and express the result using Σm notation.

10. Explain why NAND and NOR gates are known as universal gates. Show how to implement the following basic gates using only NAND gates and only NOR gates:

- (a) NOT
- (b) AND
- (c) OR